



KrishokChat: A Safety-First Agentic Agricultural Advisory System for Bangladeshi Smallholder Farmers

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CAPSTONE
2026
NSU · ECE

85,979

benchmark instances, 4 tracks

1.9×

Bengali QA over best zero-shot

95–97%

on-device crop diagnosis

<30 ms

offline Android inference

2 paisa

marginal cost per AI answer

16123

human escalation, always free

The Problem

Bangladesh has 47 million smallholder farming households with no reliable advisory in their own language. A wrong pesticide dose can harm a family and pollute local water. Standard chatbots always give a chemical dosage answer when asked, whether or not the answer is correct. Crop disease left undetected costs 20–40% of yield.

Our Approach: A Layered Trust Stack

We built a system where every answer is tied back to an official government publication. The pipeline runs in strict order:

Publications → **Knowledge Graph** → **Safety Check** → **Hybrid Retrieval** → **Bengali Generation** → **Dosage Verifier** → **Cited Answer or Human Escalation**

Each stage was added because our measurements showed the previous stage was not enough on its own. See column 4 for the failure data behind each decision.

Key Contributions

- 85,979-instance Bengali agricultural benchmark across 4 evaluation tracks, every instance citation-traced to source pages.
- 2,882-node agricultural knowledge graph from 284 official publications across 13 national research institutions.
- Register-aware hybrid retrieval that closes the colloquial/formal language gap measured at R@10 0.970 (formal) vs. 0.093 (colloquial).
- Pre-retrieval safety gate for 6 agrochemical risk categories with a 14-field dosage verifier; only 0.31% non-compliance on 323 unsafe queries.
- On-device crop disease models (1.6–21.8 MB) achieving 95–97% top-1 accuracy across 5 crops, with no internet required.
- 722-image soil-moisture dataset with paired tensiometer readings; first field-collected soil dataset for Bangladesh agricultural AI.

Real-World Impact

- Health.** Unsafe chemical queries are blocked and routed to the 16123 Krishi Call Center or 999 in every product tier — safety is never paywalled.
- Environment.** Verified dosing cuts over-application. Bangladesh pilots report up to 63% water savings and 50% urea savings.
- Access.** Standard Bengali and 6 regional dialects, voice input, offline-capable on mid-range Android phones farmers already own.

How It Compares and What Comes Next

Property	Generic AI	Ours
Answers traced to a source page	—	●
Safety gate before any retrieval	—	●
Refuses when evidence is absent	—	●
Works offline on Android	—	●
Understands Bangladeshi farm dialect	○	●

● fully present ○ partial — absent

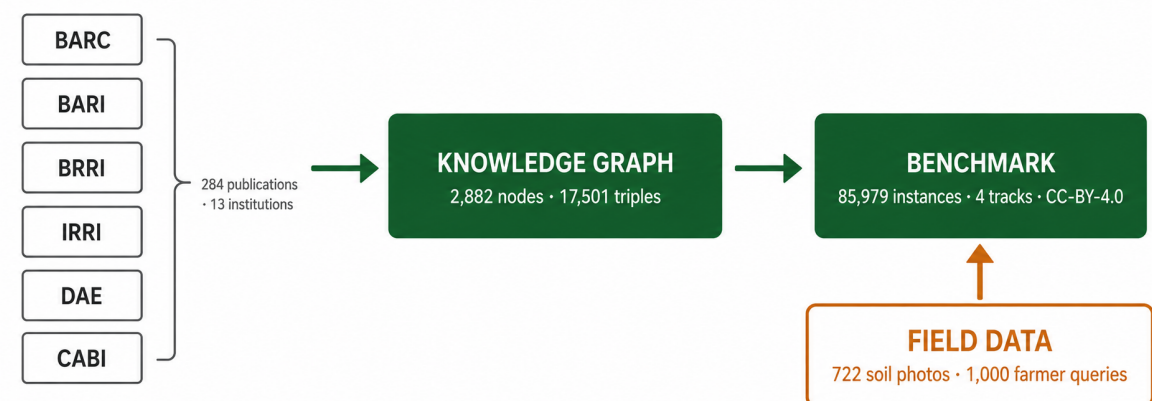
Roadmap: voice output · extended dialect coverage · soil-vision for irrigation advisories · DAE field office pilot.

Research Assets: What We Built and Released

Asset	Scale	Why it is reliable
Benchmark (4 tracks)	85,979 stances	in- General QA 28,993 · Treatment 11,224 (66.3% chemical) · Table 25,650 · Safety 20,112. Every instance traces to a source page.
Knowledge graph	2,882 nodes, 17,501 triples	19,768 entities from 284 publications at 13 institutions (BARC, BARI, BRRI, DAE and others).
Farmer queries	1,000 queries	300 face-to-face interviews in Rajshahi and Natore, covering 6 regional dialects.
Soil-moisture images	722 field photos	Tensiometer readings (0.0 to 21.5 kPa) across 14 crops, 46 series, strict train/test separation. Pabna, 2026.

All four datasets released CC-BY-4.0 on Hugging Face.

From Official Publications to Retrieval Nodes



The pipeline converts 284 official publications from 13 institutions into a structured knowledge graph (2,882 nodes, 17,501 triples). Every retrieved fact links back to its source document, making answers auditable rather than generated.

Collected in the Field, Not Scraped from the Web



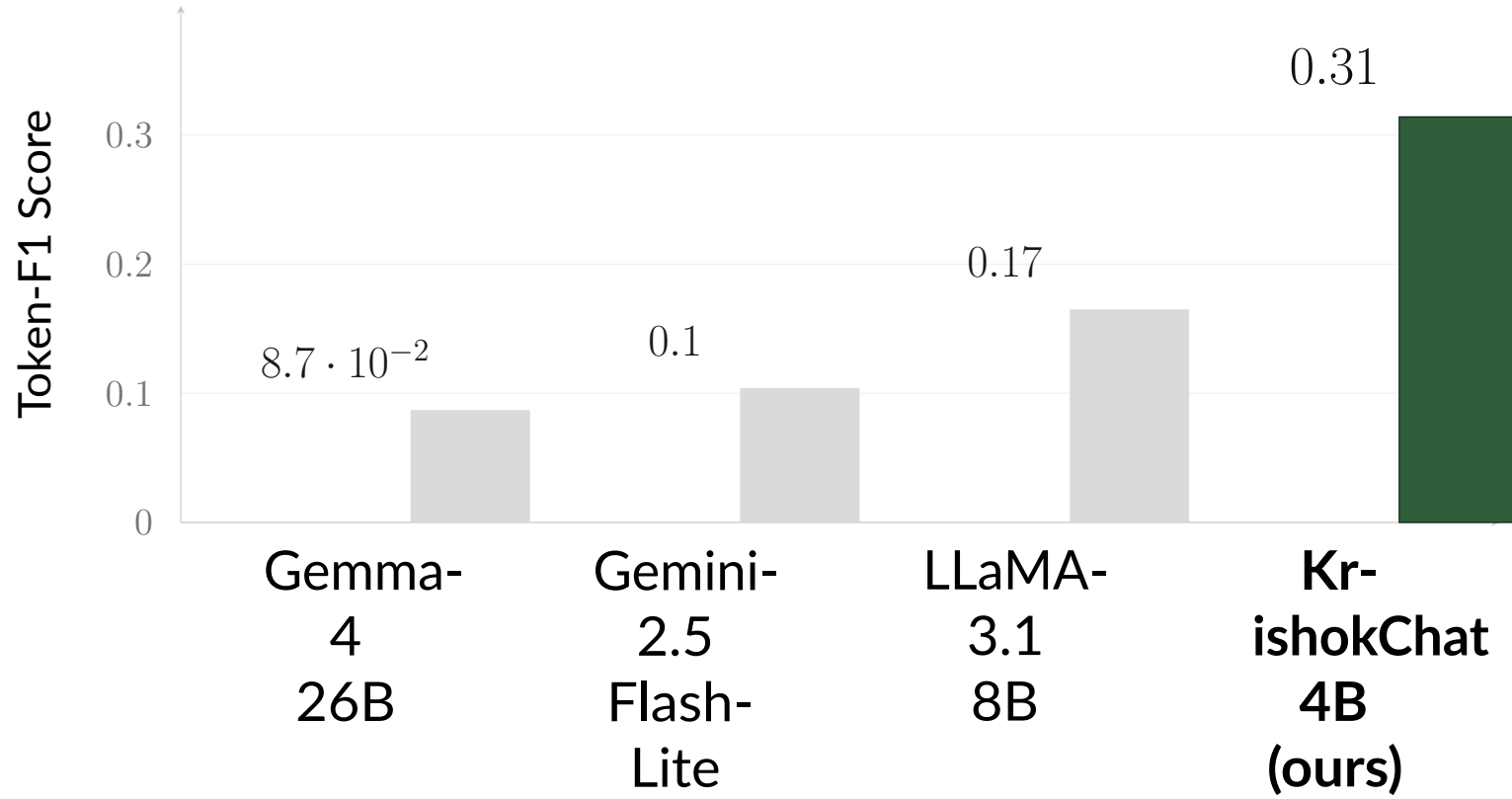
300 farmer interviews

0.0 kPa (wet)

8.0 kPa (dry)

Each soil image has a tensiometer reading as its label (physical measurement, not annotator opinion). The 300 interviews across Rajshahi and Natore captured how farmers actually describe their problems in 6 dialects.

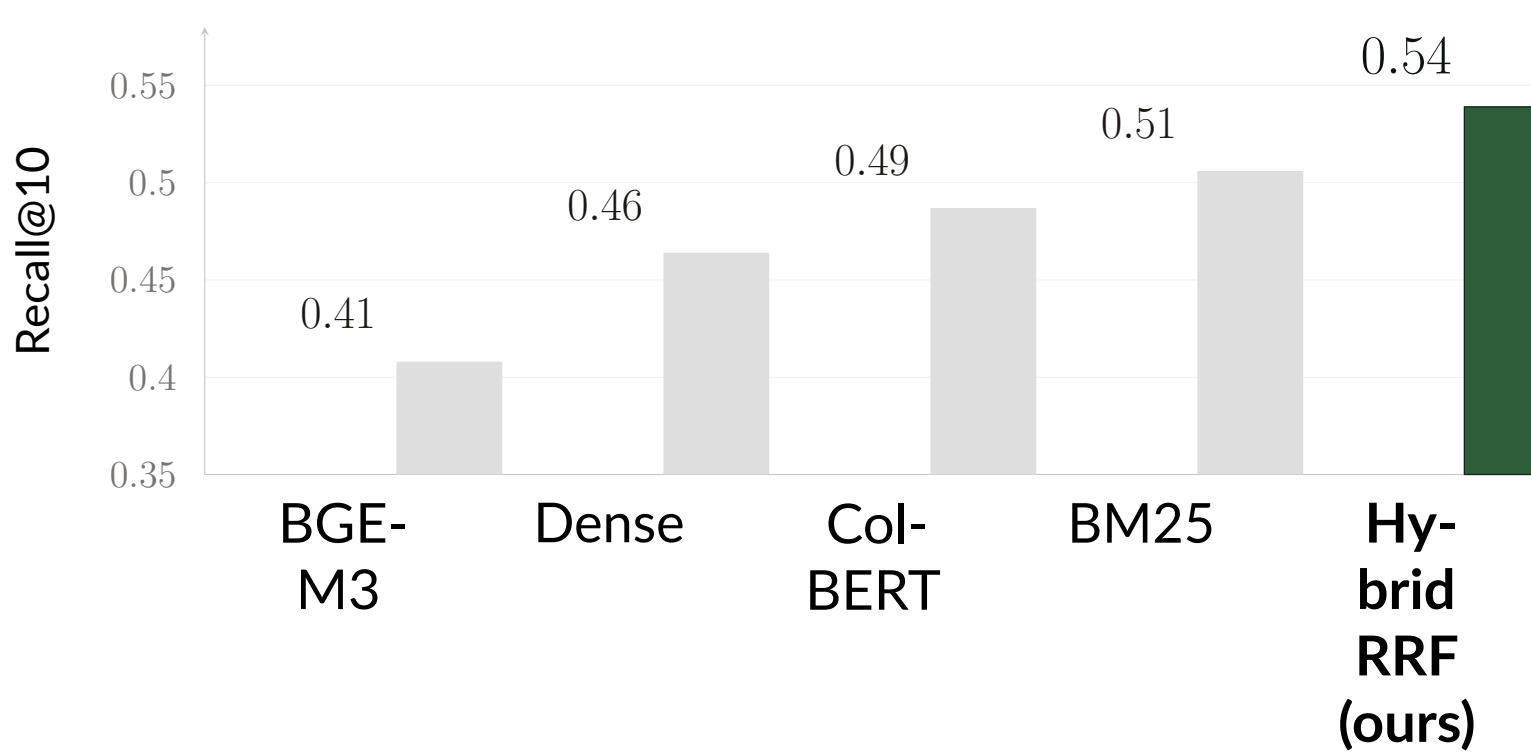
Language Model: Fine-Tuned 4B vs Larger Baselines



1.9× gain over the best zero-shot model ($p \approx 6.7 \times 10^{-5}$)

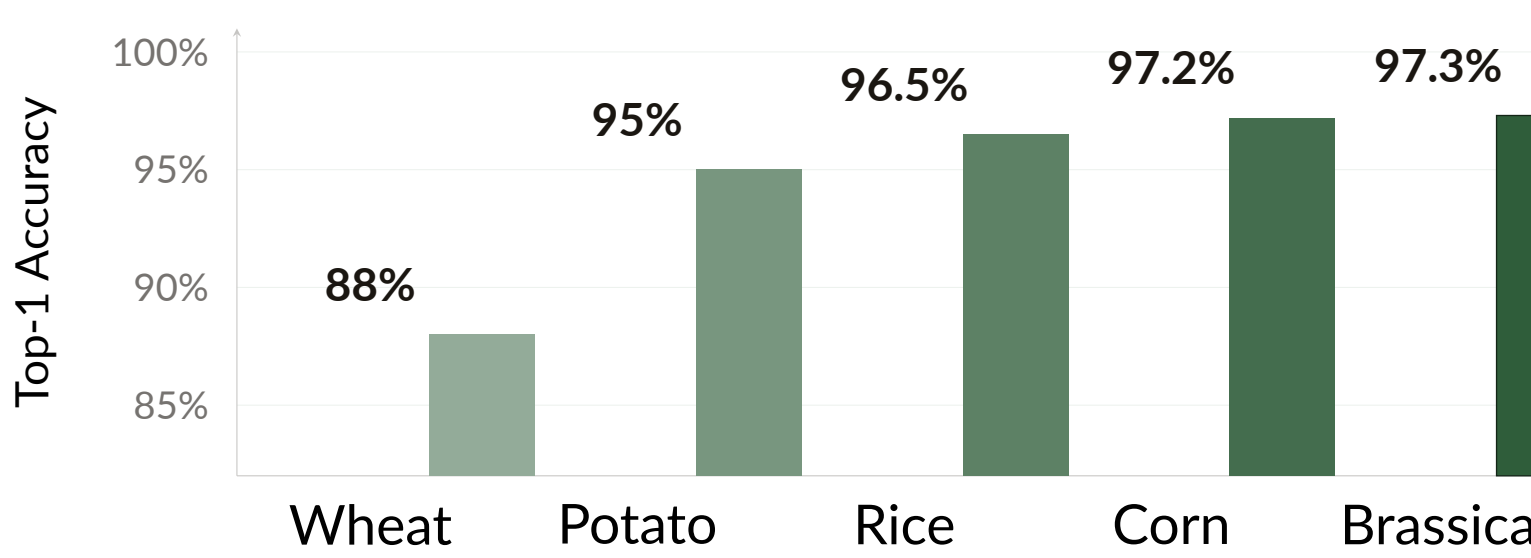
LoRA fine-tune of Gemma-4-E4B, 4-bit quantised for on-device deployment.

Retrieval: Hybrid RRF Outperforms All Methods



+9% over dense retrieval ($p < 0.001$). Dense R@10 = 0.970 (formal) vs 0.093 in colloquial farmer language.

On-Device Crop Disease Accuracy (Top-1)



436/437

Bengali treatment verified (99.8%)

<30 ms

on-device inference mid-range Android

Safety Gate: 322 of 323 Unsafe Queries Blocked

322 blocked (99.7%) 1

Out of 323 unsafe test queries, only 1 passed through. When evidence is absent, the system withholds its answer rather than guessing.

Held-out split · 900 farmer queries · per-crop test images · all on-device; no API calls.

Why Each Stage Exists: Failure Before Fix

What we measured	What we built in response
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LLMs produce wrong chemical dosages at 4.1–7.0% of queries even when the correct answer is provided directly

Dense retrieval scores R@10 of 0.970 on formal queries but drops to 0.093 on identical questions in colloquial farmer language

92% of disease terms in official documents do not appear in any form in farmer queries: পাতা হলুদ হয়ে যাচ্ছে vs. "Yellow Mosaic Virus"

The national Krishi Call Center handled 92,094 calls last year with a fixed human team

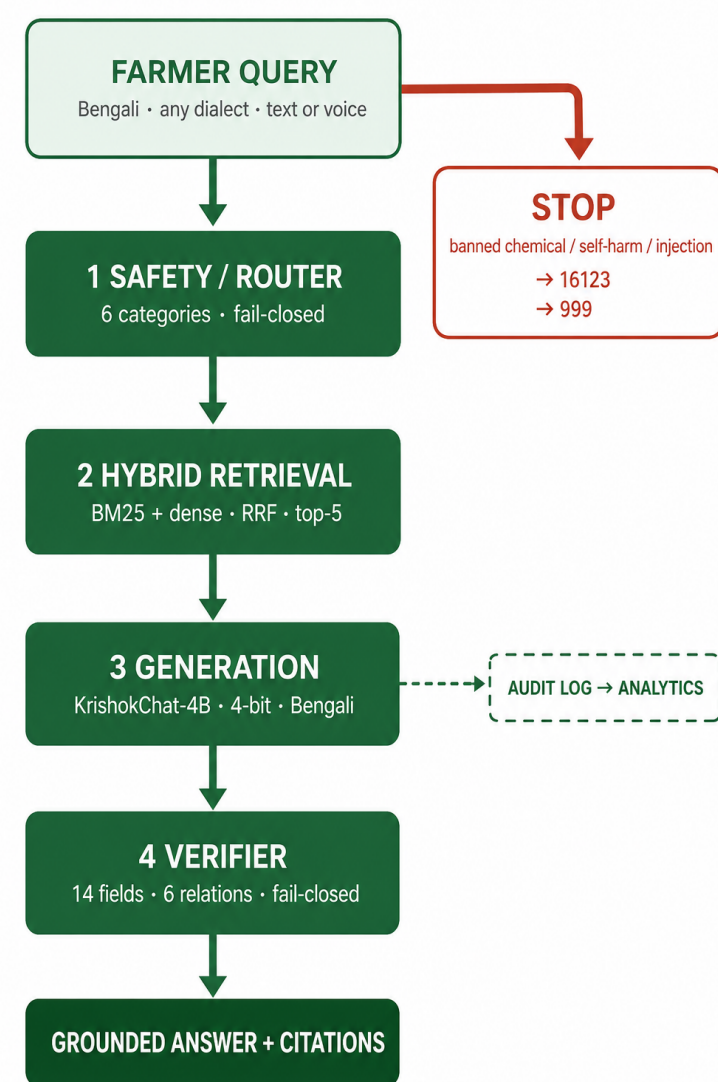
Safety check runs **before** any retrieval, plus a 14-field verifier that refuses to output if evidence is absent

Hybrid BM25 + dense retrieval fused with Reciprocal Rank Fusion

Register-aware retrieval across 6 dialects

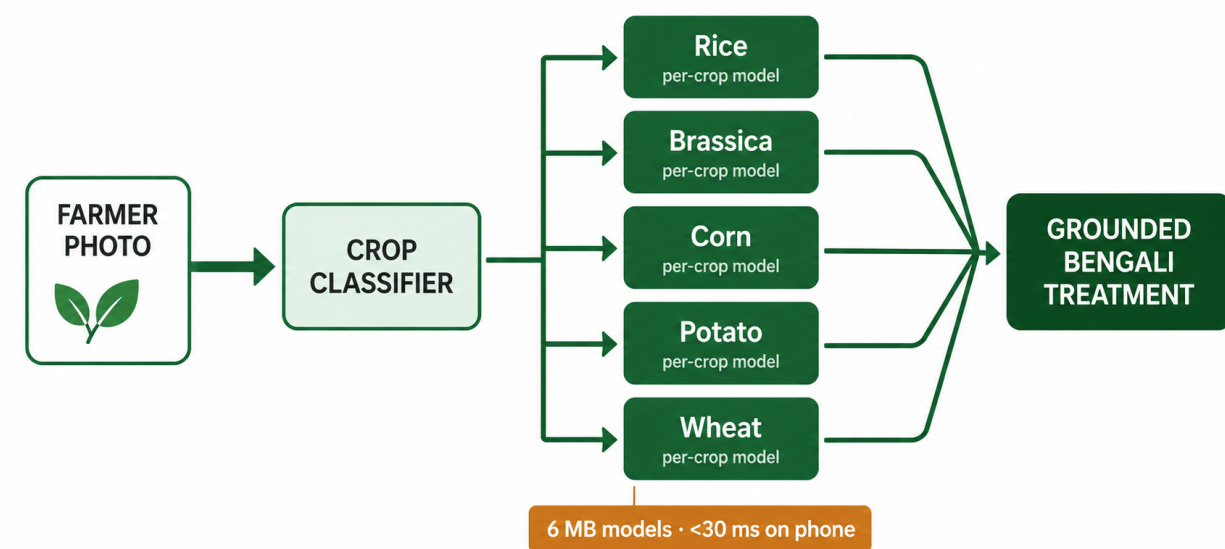
AI answers routine queries automatically; agents get only high-risk cases with a full audit trail

System Architecture: 4-Stage Safety Pipeline



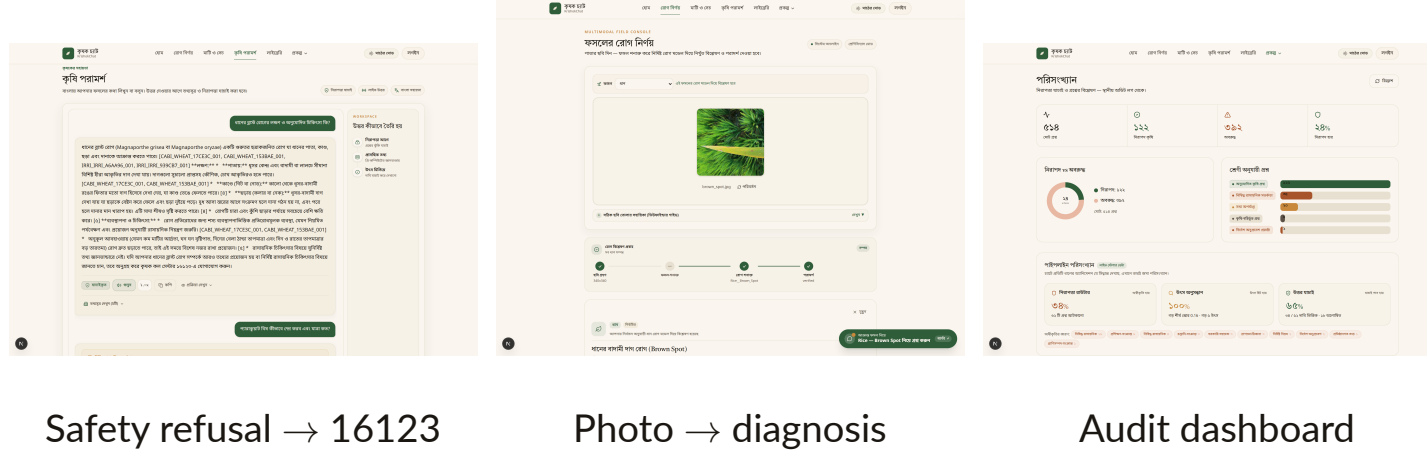
The safety router blocks any query in a high-risk category (banned chemicals, self-harm, pesticide injection) and routes it to the 16123 Krishi Call Center or 999 before any retrieval runs. Safe queries proceed through hybrid retrieval, generation, and the 14-field verifier before reaching the farmer.

On-Device Disease Classification and Routing



436/437 verified Bengali treatments. Models run fully on-device; no server, no network. Inference under 30 ms on a mid-range Android phone.

The Product: Krishi Patrak (কৃষি পত্রক)



A Bengali-first web app and Android app. Every query includes a live agent trace showing the retrieval and verification steps. Supports voice input, a high-contrast sunlight mode, and works with no internet connection.

Market Readiness

What is working now. Full-stack web and Android app, on-device crop disease models, live safety pipeline with an audit dashboard, and a 437-entry verified treatment library tested end-to-end.

The offline case. 73.4% of Bangladeshi households own a smart-phone, but only 43.6% of rural residents have internet access. A farming advisory that requires connectivity reaches less than half the intended audience.

Unit economics. Each AI answer costs roughly 2 paisa on a shared API. At scale, a single GPU server at \$497/month handles around 13 million answers per month.

Business Model: Farmers Use It Free; Institutions Pay

FREE — EVERY FARMER	PREMIUM — INSTITUTIONS & PRO
✓ Grounded answers — no LLM ✓ On-device photo diagnosis ✓ Safety gate + 16123 ✓ Works offline	★ Everything in Free ★ Open-ended AI advisory ★ Priority + voice
Safety gate & 16123 escalation — identical in both tiers	

B2G. The system acts as an AI front desk for the Krishi Call Center. Routine queries are answered automatically. High-risk cases go to a human agent with a complete audit trail that satisfies DAE and a2i reporting requirements.

B2B. An analytics dashboard gives agro-dealers, seed companies, and NGOs district-level data on what farmers are asking and which crop diseases are rising in a given season.

Data licensing. The 85,979-instance benchmark and 722-image soil dataset are CC-BY-4.0 for academic use and available under a commercial licence for agri-fintech and crop insurance applications.

Papers, Code and Data



Dataset (Hugging Face)



Code (GitHub)

Full references listed in the footer below.

[1] KrishokChat: A Provenance-Traceable Multi-Task Bengali Agricultural Benchmark with Safety-Critical Chemical Advisory. *EACL 2026, under review.*

[2] AgriTrust: KG-Grounded Register-Aware Retrieval for Bengali Agricultural Advisory. *SIGIR-AP 2026, under review.*

[3] AgriVision BD: On-Device Crop Disease Classification. *Technical Report, 2026.*